Separating the Nonzero Part of a Finite Algebra

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Abstract

We give an explicit polynomial, using one additional rational coordinate, for a guarded rational point of a finite algebra obtained by adjoining quadratic roots to an algebra with separable fibers. A characteristic polynomial identifies the factors on which the guard is nonzero. The product of its nonzero eigenvalues then bounds every required quadratic root separation; only the scalar discriminants of the initial separable polynomials are needed. The construction covers degree drops, nonreal points, and nonreduced fibers where the guard vanishes. Applied to the specified arithmetic sources, it gives a family in 27 total variables of degree less than 2^{87} and a fully specified polynomial in 29 total variables of degree less than 2^{89} and coefficient height at most $2^{2^{99}}$. The elimination theorem is unconditional. Assertions about defining the integers retain the arithmetic premises and the effective or ineffective thresholds of the respective sources.

1 The algebra and its guarded rational points

Let $R_0 = \mathbb{Z}[x_1, \dots, x_a]$ and consider the finite free algebras

$$B = R_0[z_1, \dots, z_p]/(f_1(z_1), \dots, f_p(z_p)), \qquad A = B[q_1, \dots, q_s]/(q_j^2 - c_j)_{j=1}^s.$$

Here $p, s \geq 1$, the polynomials $f_i \in R_0[Z]$ are monic of degrees $d_i \geq 2$, and $c_j \in B$. Assume that every f_i is separable at every rational specialization of the base coordinates. Write $m = p + s$ and $\theta_1, \dots, \theta_m$ for the generators in this order, with $d_{p+j} = 2$. Put

$$b = \prod_{i=1}^{p} d_i, \qquad n = b2^s.$$

The monomials with each exponent below its defining degree are bases of B and A. For an element u, let M_u denote its multiplication matrix in the indicated basis.

Fix $g \in B$ and put

$$G = gq_1 \cdots q_s, \qquad T = g^2 c_1 \cdots c_s \in B.$$

Thus $G^2 = T$ in A. We encode, at a rational base point,

$$\text{there is a homomorphism } A \longrightarrow \mathbb{Q} \text{ over that base point with } G \neq 0. \tag{1}$$

There are no further residual equations. Every algebra generator will be eliminated; precisely one new rational coordinate U is used. At a rational base point B is an étale $\mathbb{Q}$-algebra. On its factors where $T \neq 0$, all c_j and g are nonzero, so the corresponding part of A is also étale. The remaining part of A may be nonreduced.

2 An explicit integer polynomial

Define scalar integer polynomials

$$\det_B(I - vM_T) = \sum_{j=0}^{b} a_j v^j, \quad a_0 = 1, \qquad \delta_i = \mathrm{Disc}(f_i) \quad (1 \le i \le p).$$

Use the convention $\mathrm{Disc}(f) = (-1)^{d(d-1)/2}\,\mathrm{Res}(f, f')$ for a monic polynomial of degree d. Define

$$R = n\left(1 + \sum_{i=1}^{m}\sum_{u,v=1}^{n} (M_{\theta_i}^A)_{uv}^2 + \sum_{u,v=1}^{b} (M_g^B)_{uv}^2\right), \qquad \sigma = \sum_{i=1}^{p} d_i(d_i - 1). \tag{2}$$

The superscripts specify the algebra in which multiplication is taken.

For each integer $1 \le r \le b$, use only the following polynomial operations:

$$c = a_r, \qquad h_r(Z) = \sum_{j=0}^{r} a_j Z^{r-j}, \qquad E = c - h_r(T) \in B,$$

$$D = c^2 \prod_{i=1}^{p} \delta_i^2, \qquad S_r = r(s+1) + \sigma,$$

$$K = (2R)^{S_r}(1 + c^2)\prod_{i=1}^{p}(1 + \delta_i^2), \qquad C = D + 4RK.$$

These symbols depend on r. Set

$$\beta_r = DE\left(C^m + \sum_{i=1}^{m} D^i C^{m-i}\theta_i\right) \in A, \qquad H_r(\mathbf{x}, U) = \det_A(I - UM_{\beta_r}). \tag{3}$$

Finally define

$$Q_r = \begin{cases} H_r^2 + \displaystyle\sum_{j=r+1}^{b} a_j^2, & r < b, \\ H_b, & r = b, \end{cases} \qquad P(\mathbf{x}, U) = \prod_{r=1}^{b} Q_r. \tag{4}$$

All finite index ranges and integer coefficient operations are explicit. No division occurs; the expressions remain polynomials where $c = 0$.

Theorem 2.1. *For every rational base point* $\mathbf{x}$*, predicate* (1) *holds if and only if there is a rational* U *satisfying* $P(\mathbf{x}, U) = 0$*.*

3 Proof of the projection theorem

Fix a rational base point throughout this section. A geometric character means an algebra homomorphism to $\overline{\mathbb{Q}}$ over $\mathbb{Q}$. Let r_0 be the number of characters χ of B with $T(\chi) \neq 0$, counted individually.

Lemma 3.1. *The degree of* $\det_B(I - vM_T)$ *is* r_0*. For* $r = r_0 > 0$*, one has* $c \neq 0$*, and* E/c *equals one on the factors with* $T \neq 0$ *and zero on all remaining factors.*

Proof. Since B is étale, the determinant is $\prod_\chi (1 - vT(\chi))$. For $r = r_0$, the polynomial h_r is the product of $Z - T(\chi)$ over those characters with $T(\chi) \neq 0$, including repetitions of their values. It vanishes at every such value and has constant term c. Thus $E = c - h_r(T)$ has the stated values. □

For the correct case $r = r_0 > 0$, all scalar δ_i are nonzero by hypothesis, so $D > 0$, $K > 0$, and

$$\varepsilon = D/K = \frac{c^2}{1+c^2} \prod_{i=1}^{p} \frac{\delta_i^2}{1+\delta_i^2} (2R)^{-S_r} \in (0, 1].$$

Lemma 3.2. *In the correct case, two characters of A with $G \neq 0$ which first differ at θ_i differ there in complex absolute value by at least ε. The element*

$$\alpha = L^m + \sum_{i=1}^{m} L^{m-i}\theta_i, \qquad L = C/D = 1 + 4R/\varepsilon,$$

has distinct nonzero values on all these characters.

Proof. Every value of every θ_i, and every value of g, has absolute value at most R. Indeed these values are eigenvalues of the respective multiplication matrices. The spectral radius of a k-by-k matrix is at most k times the maximum absolute entry; $k \leq n$ and $|t| \leq 1 + t^2$ prove the assertion from (2). In particular $R \geq 1$.

First suppose $i \leq p$, and let Δ be the difference in the ith generator. The discriminant δ_i is the product of the squared pairwise root differences of f_i. Every root has absolute value at most R. With $k_i = d_i(d_i - 1)$, this gives

$$|\Delta| \geq \frac{\sqrt{|\delta_i|}}{(2R)^{(k_i-2)/2}}.$$

For every real $z > 0$ one has $\min(1, z^2) \leq \sqrt{z}$. Consequently

$$\frac{\delta_i^2}{1+\delta_i^2}(2R)^{-k_i} \leq \frac{\sqrt{|\delta_i|}}{(2R)^{(k_i-2)/2}}.$$

The other factors in ε are at most one, and $S_r \geq k_i$. Hence $\varepsilon \leq |\Delta|$.

Next suppose $i = p + j$. Since the earlier point generators agree, the two characters lie above the same character χ of B, and their values are the two roots of $Z^2 - c_j(\chi)$. Now

$$|c| = \prod_{\psi:\, T(\psi)\neq 0} |g(\psi)|^2 \prod_{k=1}^{s} |c_k(\psi)|.$$

There are $r(s+1)$ factors in this product; every factor is at most R^2. Isolating the particular nonzero factor $|c_j(\chi)|$ yields

$$|c_j(\chi)| \geq |c| R^{-2r(s+1)+2}, \qquad |\Delta| = 2\sqrt{|c_j(\chi)|} \geq 2\sqrt{|c|} R^{1-r(s+1)}.$$

On the other hand,

$$\varepsilon \leq \frac{c^2}{1+c^2}(2R)^{-r(s+1)} \leq \sqrt{|c|}(2R)^{-r(s+1)} \leq |\Delta|.$$

This proves the separation estimate, also for complex roots. It uses the product of the nonzero values of T, without requiring those values to be pairwise distinct.

For two characters first differing at θ_i, the leading term in their α-difference has absolute value at least εL^{m-i}. The remaining terms have total absolute value less than

$$\frac{2R}{L-1} L^{m-i} = \frac{\varepsilon}{2} L^{m-i}.$$

The values of α are therefore distinct. Also, the sum after L^m has absolute value less than $RL^m/(L-1) \leq L^m/4$, so none of its values is zero. □

Proof of Theorem 2.1. If $c = 0$ in a case, then $D = 0$, $\beta_r = 0$, and $H_r = 1$. For $r < b$, a rational zero of Q_r therefore forces $c \neq 0$ and $a_j = 0$ for every $j > r$; it also forces $H_r = 0$ by the sum-of-squares identity. For $r = b$, a zero of H_b similarly forces $c = a_b \neq 0$. Thus any zero uses precisely the correct positive degree case.

By Lemma 3.1, multiplication by E vanishes identically on the part of A above $T = 0$, including its nilpotents. On the other part one has

$$\beta_r = cD^{m+1}\alpha.$$

This part is étale. Lemma 3.2 implies that its β_r-values are distinct and nonzero. Hence H_r is the product of the distinct factors $1 - U\beta_r(\chi)$ over the characters with $G \neq 0$. A rational zero gives a one-dimensional rational kernel of $I - UM_{\beta_r}$. All native multiplication matrices commute with M_{β_r} and preserve this line. Their actions on it are rational scalars satisfying the algebra relations. They yield a rational character. Its β_r-value is nonzero, so it has $G \neq 0$. This proves soundness.

Conversely a guarded rational character implies $r_0 > 0$. In the correct case its β_r-value is nonzero and rational. Its reciprocal annihilates H_r, and all tail coefficients a_j, $j > r$, vanish. It therefore annihilates Q_r and P. If $r_0 = 0$, every $c = a_r$ is zero and every H_r is one, so no factor has a zero. This includes the empty guarded part. □

4 Degree and coefficient estimates

Give each base coordinate weight one, and give the native generators weights $w_i \geq 1$ for which monic reduction does not increase weight. For coefficient norms use the weighted column norm, with basis scaling $2^{\sum e_i\gamma_i}$ on the monomial $\prod \theta_i^{e_i}$. Scalar polynomials have the coefficient ℓ^1 norm. Suppose the following bounds are supplied:

$$\|M_{\theta_i}\| \leq 2^{\gamma_i}, \quad \operatorname{wt}(G) \leq w_G, \quad \|M_G\| \leq 2^{\eta}, \quad \operatorname{wt}(g) \leq w_g, \quad \|M_g^B\| \leq 2^{\eta_g}.$$

The γ_i, η, η_g are nonnegative integers. The companion-column recurrence in [3] supplies these bounds from the integer coefficient circuit. The B-basis uses the restrictions of the same weights. Since $G^2 = T \in B$, its weight and logarithmic operator norm are bounded by $t = 2w_G$ and $\mu = 2\eta$. For the norm assertion, B is an invariant subspace of multiplication by G^2.

For positive integers u, write $\ell(u) = \lceil \log_2 u \rceil$. Set

$$\begin{aligned}
w_* &= \max\left\{\sum_{i=1}^{m}(d_i-1)w_i + \max_i w_i, \sum_{i=1}^{p}(d_i-1)w_i + w_g\right\},\\
\gamma_* &= \max\left\{\sum_{i=1}^{m}(d_i-1)\gamma_i + \max_i \gamma_i, \sum_{i=1}^{p}(d_i-1)\gamma_i + \eta_g\right\},\\
R_d &= 2w_*, \qquad R_l = \ell(n) + \ell(1+mn^2+b^2) + 2\gamma_*, \qquad a_l = \mu + \ell(b),\\
u_i &= d_i(d_i-1)w_i, \qquad v_i = d_i(d_i-1)\gamma_i + \ell\big((2d_i-1)!d_i^{d_i}\big) \quad (i \leq p),\\
u &= \sum_{i=1}^{p} u_i, \qquad v = \sum_{i=1}^{p} v_i.
\end{aligned}$$

Define four integer majorants

$$\begin{aligned}
W_1 &= (3+2m)t + m(s+1)R_d,\\
W_0 &= 2(m+1)u + m(\sigma+1)R_d + \max_i w_i,\\
L_1 &= (3+2m)a_l + m(s+1)(1+R_l),\\
L_0 &= 2(m+1)v + \ell(b) + \ell(m+1)\\
&\qquad + m\big(\sigma(1+R_l) + p + 4 + R_l\big) + \max_i \gamma_i.
\end{aligned}$$

The letters L_0, L_1 here denote coefficient bounds and are unrelated to the fixed coefficient exponent in the applications.

Theorem 4.1. *The polynomial in (4) satisfies*

$$\deg P \le n(W_1 b^2 + (W_0+1)(2b-1)), \tag{5}$$

$$\log_2 \|P\|_1 \le n(L_1 b^2 + (L_0 + \ell(n) + 1)(2b-1)) + (b-1)\ell(b+1). \tag{6}$$

Its coefficient height is at most $\|P\|_1$.

Proof. An entry in the multiplication matrix of an element of weight w has degree at most $w+$ column basis weight $-$ row basis weight. The analogous logarithmic coefficient-norm estimate uses the γ weights. This proves the bounds R_d, R_l for (2).

In every principal minor the row and column weights cancel. If multiplication by an element of an algebra of rank h has weight at most w and operator norm at most 2^λ, then the jth characteristic coefficient has degree at most jw and norm at most $\binom{h}{j} 2^{j\lambda} \le 2^{j(\lambda+\ell(h))}$. For the norm estimate, apply the diagonal similarity specifying the weighted column norm; the column-norm determinant estimate bounds each principal minor. In particular

$$\deg a_j \le jt, \qquad \log_2 \|a_j\|_1 \le j a_l.$$

The constant term cancels in $E = c - h_r(T)$, leaving exactly r terms, each of weight at most rt and operator norm at most 2^{ra_l}. Thus

$$\mathrm{wt}(E) \le rt, \qquad \log_2 \|M_E\| \le r a_l + \ell(b).$$

The discriminant is weighted homogeneous of weight $d_i(d_i-1)$ in its roots. Its Sylvester determinant has at most $(2d_i-1)!$ terms, each with at most d_i derivative coefficients, bounded in addition by d_i. For a scalar native polynomial, the weighted last column of its companion matrix gives $\|[Z^j]f_i\|_1 \le 2^{(d_i-j)\gamma_i}$ and $\deg[Z^j]f_i \le (d_i-j)w_i$ for $j < d_i$. The Sylvester expansion with these coefficient bounds gives degree at most u_i and logarithmic norm at most v_i.

Products and sums now yield

$$\begin{aligned}
&\deg D \le 2rt + 2u, && \log_2 \|D\|_1 \le 2ra_l + 2v,\\
&\deg K \le S_r R_d + 2rt + 2u, && \log_2 \|K\|_1 \le S_r(1+R_l) + 2ra_l + 2v + p + 1,\\
&\deg C \le (S_r+1)R_d + 2rt + 2u, && \log_2 \|C\|_1 \le S_r(1+R_l) + 2ra_l + 2v + p + 4 + R_l.
\end{aligned}$$

These bounds for C dominate the corresponding bounds for D. Applying them to (3) gives

$$\mathrm{wt}(\beta_r) \le W_1 r + W_0, \qquad \log_2 \|M_{\beta_r}\| \le L_1 r + L_0.$$

The principal-minor estimate applied to H_r gives

$$\deg H_r \le n(W_1 r + W_0 + 1), \qquad \log_2 \|H_r\|_1 \le n(L_1 r + L_0 + \ell(n) + 1).$$

Both estimates dominate those for all tail coefficients a_j in Q_r: one may directly compare with bt and ba_l, using $n \ge b$, $r \ge 1$, and the displayed positive majorants. For $r < b$, square and sum at most $b+1$ terms; for $r = b$, keep the bound for H_b without squaring. Sum the bounds over the factors and use

$$2\sum_{r=1}^{b-1} r + b = b^2, \qquad 2\sum_{r=1}^{b-1} 1 + 1 = 2b - 1.$$

This proves (5) and (6). □

5 The arithmetic sources

The sources in [1, 2] retain respectively 26 and 28 rational base coordinates, including the distinguished input. Each has nine independent scalar monic quartics for the point charts, followed by three or four monic quadratic relations. Every quadratic radicand belongs to the point algebra B. Each pole generator occurs with exponent exactly one in the complete guard, whose remaining factor g belongs to B. There are no residual equations or additional indexed guards. These assertions are checked on every coefficient dependency in the supplied circuits.

For completeness, the separability of the point charts holds at *every* rational base point. For rational h_0, h_1, put

$$A_0 = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \quad B_0 = 4h_0h_1(h_0^2 + 2h_1^2) + 6,$$

$$U_0 = 6A_0^3 - 9A_0B_0^2 - 4B_0^3, \quad V_0 = -4A_0^3 + 6A_0B_0^2 + 3B_0^3.$$

The point polynomial is $f(Z) = Z^4 - 4U_0Z^2 + 8V_0^2$, with discriminant

$$2^{15}V_0^2(2A_0^2 - B_0^2)^4(A_0^2 - 2B_0^2)^2.$$

The pair (A_0, B_0) cannot vanish: otherwise $(h_0 + h_1\sqrt{2})^4 = (3 - \sqrt{2})^2$, whose norm would make 49 a rational fourth power. Neither quadratic norm factor can vanish at a nonzero rational pair. Finally $V_0 = 0$ would produce a rational root of $4X^3 - 6X - 3$, which is Eisenstein at 3. Thus all hypotheses of Theorem 2.1 hold on every rational base point; no branchwise guard is replaced by the stronger assertion that its algebra norm is nonzero.

Quantity	27 variables	29 variables
m	12	13
b	2^{18}	2^{18}
n	2^{21}	2^{22}
w_G	4 423 559	6 789 625
w_g	803 872	1 234 707
η at exponent 137731	6 998 669 574	7 005 250 004
η_g at exponent 137731	1 106 350 691	1 107 547 330
Number of factors	262 144	262 144

The nine point generator weights are 6 and their logarithmic operator bounds are 14. The remaining weights are

$$(1568138, 1935231, 116318) \quad \text{for 27 variables,}$$

$$(1568138, 1935231, 1935231, 116318) \quad \text{for 29 variables.}$$

Every logarithmic generator bound and integer recurrence is recorded in the accompanying coefficient specifications.

Corollary 5.1. *Under the arithmetic premises of the 27-variable source, the construction gives a computable family in 27 total variables such that every sufficiently large odd fixed coefficient exponent defines $\mathbb{Z}$ over $\mathbb{Q}$, and*

$$\deg P \le 1112959446302114713614417 92 < 2^{87}.$$

The sufficient coefficient threshold is ineffective.

Proof. Theorem 2.1 preserves the source projection exactly. The native degree weights are independent of the fixed coefficient exponent. Substitute them into (5) and apply the source arithmetic theorem, with its unchanged threshold. □

Corollary 5.2. *Under the arithmetic premises of the explicit 29-variable source, the construction at fixed exponent* 137731 *defines* $\mathbb{Z}$ *over* $\mathbb{Q}$ *and satisfies*

$$\deg P \le 394212872202031272474705920 < 2^{89},$$

$$\log_2 H(P) \le 55857834363941215765022402148 5 < 2^{99}.$$

Proof. Apply Theorem 2.1 to the effective source and substitute its integer data into (5) and (6). No arithmetic parameter is enlarged. □

At the same numerical exponent the 27-variable member satisfies $\log_2 H(P) \le 217401420128398649581869203437 < 2^{98}$. That exponent is an algebraic reproduction fixture for this family; it is not certified to exceed the ineffective defining threshold. Finite algebraic verification does not establish a source arithmetic premise or make that threshold effective.

6 Exact specification and verification

The program `build_active_specification.py` reads the pinned integer coefficient circuits, verifies the guard and dependency assertions, recomputes the weighted companion bounds, and writes the formulas (2)–(4) with their explicit finite ranges. The separate checker reconstructs the coefficient dependencies and integer majorants without importing the builder. It sums the degree and coefficient estimates over all 262144 cases literally, and checks the source and output hashes and the precise effective or ineffective interpretation.

The independent program `check_active_degenerations.py` checks small exact algebras with repeated nonzero T-values, inactive nilpotents, nonreal roots, no rational characters, and an empty guarded part. It computes each characteristic polynomial both by determinant expansion and by Newton traces, checks the idempotent identity, and verifies squarefreeness by an exact polynomial gcd. These examples test degeneracies; the proof of the general projection theorem is given above.

The rank-sized matrices and full monomial list are not materialized. An expression with determinants, coefficient extraction, finite integer sums and products specifies every coefficient, without introducing any existential coordinate for an internal index or matrix entry. All verification programs use the Python standard library.

References

[1] *A defining polynomial in twenty-seven variables: an existence theorem with a uniform degree bound*, companion paper and guarded source in the accompanying repository, `papers/few_variables27`.

[2] *An explicit twenty-nine-variable polynomial definition of the integers*, companion paper and guarded source in the accompanying repository, `papers/few_variables`.

[3] *Weighted bounds for eliminating finite algebraic choices*, the companion-column and principal-minor bounds, companion paper in `papers/finite_algebra`.